Imaging the Child's Bowel: Intestinal Ultrasound, Magnetic Resonance Enterography, Computed Tomography and the Upper Gastrointestinal Contrast Study.

Imagine four children who need a picture of their bowel on the same day. The first is a fourteen-year-old with Crohn's disease of the terminal ileum who is about to start an anti tumor necrosis factor drug, and his family wants to know, early, whether it is working. The second is a twelve-year-old with weight loss, mouth ulcers and a raised fecal calprotectin, who has just had an ileocolonoscopy. The third is a four-year-old with bloody diarrhea whose ultrasound report says the colon wall measures two point three millimeters, and is therefore normal. The fourth is a ten-day-old baby who has vomited green bile three times. Each of these children needs a different picture, and in each one the wrong picture, or the right picture read with the wrong numbers, can lead to harm.

The clinical question of this chapter is which imaging test to choose for which child, and how to read it. The main idea is this. A child's bowel wall is thinner than an adult's, so adult cut-offs miss disease. Intestinal ultrasound is the first and repeatable look, needs no radiation and no sedation, and can show within weeks whether a biologic is working. Magnetic resonance enterography is the map of the small bowel at diagnosis and the measure of deep healing, and in children it can now be scored with a dedicated index that needs no intravenous contrast. Computed tomography carries a small but real cumulative cancer risk and is reserved for emergencies. And for the baby with bilious vomiting, ultrasound has caught up with the upper gastrointestinal contrast study, but the contrast study still decides when the ultrasound is unclear.

We will move through the topic in order. First, what inflammation looks like on an image. Second, intestinal ultrasound: normal values, accuracy, scores and early response. Third, magnetic resonance enterography and the pediatric index from the ImageKids study. Fourth, computed tomography and radiation. Fifth, malrotation and the infant with bilious vomiting. Sixth, intussusception, appendicitis and Hirschsprung disease. We then compare the guidelines, list the traps, and finish with what to do differently in clinic.

A note on sources. For normal values and accuracy of ultrasound we use the systematic reviews by van Wassenaer and colleagues, published in 2019 in the Journal of Crohn's and Colitis and in 2020 in Pediatric Radiology, a 2026 study by Räisänen and colleagues in Inflammatory Bowel Diseases, and the 2023 expert consensus of the pediatric committee of the International Bowel Ultrasound Group, led by Kellar. For early response we use the Super

Sonic study by Dolinger and colleagues, published in 2024. For magnetic resonance we use the ImageKids study by Focht and colleagues, published in Gastroenterology in 2022. For radiation we use the British cohort of Pearce and colleagues in the Lancet in 2012, the Australian cohort of Mathews and colleagues in the British Medical Journal in 2013, and the European EPI-CT studies published in 2023. For malrotation we use the series of Sizemore and colleagues from 2008 and the 2021 meta-analysis by Nguyen and colleagues. For guidelines we use the revised Porto criteria of the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, the 2025 joint European guideline of the European Crohn's and Colitis Organisation, known as ECCO, with the radiology and ultrasound societies, the 2022 Appropriateness Criteria of the American College of Radiology, and the 2025 Crohn's disease guideline of the American College of Gastroenterology.

Part one. What inflammation looks like on an image.

Crohn's disease is a transmural disease. It inflames the whole thickness of the bowel wall and spreads into the fat and mesentery around it. Every imaging sign follows from that. The first and most reproducible sign is thickening: edema and inflammatory cells widen the wall. The second is hyperemia, an increase in blood flow. On ultrasound it is seen with color Doppler and graded by the Limberg score from zero to four; on magnetic resonance it is seen as increased enhancement after intravenous gadolinium. The third is loss of stratification: the normal layers of the wall, which on ultrasound alternate as bright and dark lines, blur or disappear, and ulcers break the mucosal line. The fourth is change in the mesentery: inflamed fat that wraps around the bowel, engorged mesenteric vessels lined up like the teeth of a comb, called the comb sign, and mesenteric edema. The fifth applies to magnetic resonance only and is restricted diffusion: on diffusion weighted imaging, water moves less freely in densely inflamed tissue, and the wall looks bright.

Two cautions follow. First, these signs describe activity, which is different from damage. A fixed narrowing with dilated bowel upstream is a stricture, and the 2018 consensus of the Society of Abdominal Radiology and the Society for Pediatric Radiology, led by Bruining, reserves the word stricture for a narrowing with upstream dilatation. Second, restricted diffusion on its own should not be used to call disease active, because normal collapsed jejunum also restricts diffusion. Diffusion must be read together with wall thickness, ulcers and the mesentery.

Part two. Intestinal ultrasound.

Start with what is normal. In the 2020 review by van Wassenaer and colleagues, which pooled seven studies of about 207 healthy children, the upper limit of normal bowel wall thickness

was one point nine millimeters, for both the small bowel and the colon. The jejunum was thinner, with an upper limit around one point one millimeters, and the ileum and cecum reached one point nine. Children under four had a slightly thicker ileum, by about zero point four millimeters, and the colon thickened by a few tenths of a millimeter through adolescence, a change the authors considered clinically unimportant. The key point is the comparison with adults. Adult practice usually calls a wall abnormal above three millimeters. In a child, three millimeters is well above the normal range, so an adult cut-off will miss disease.

How much does it miss? In the 2026 study by Räisänen and colleagues, 144 ultrasound examinations in 133 children were compared with colonoscopy. A colon wall of two point seven millimeters or more, in children aged six and over, had a sensitivity of sixty-seven percent and a specificity of eighty-eight percent, and the sensitivity rose to eighty-five percent for moderate to severe disease. In the terminal ileum, the same cut-off had a sensitivity of sixty-one percent and a specificity of seventy-six percent. Most strikingly, in very early onset inflammatory bowel disease, meaning disease starting before six years of age, the adult cut-off of three millimeters missed sixty percent of inflamed ilea, and the authors proposed a tentative cut-off of two point two millimeters for that age group. An Italian study by Dell'Era and colleagues, using a cut-off above three millimeters in all segments, found a sensitivity of about seventy percent and a specificity of about ninety-seven percent. Across fourteen pediatric studies in the 2019 review, sensitivity at diagnosis ranged widely, from thirty-nine to ninety-three percent, while specificity ranged from ninety to one hundred percent. Put simply, a clearly thickened, hyperemic wall in a child is very likely to be inflamed, but a normal looking scan does not rule disease out.

To turn measurements into a score, van Wassenaer and colleagues developed the pediatric Crohn disease intestinal ultrasound score in seventy-four children, scoring sixty-seven terminal ileal and 364 colonic segments against the endoscopic score called SES-CD. In the terminal ileum, a wall of two to three millimeters scored one point, three to three point seven scored two, and more than three point seven scored three. In the colon the thresholds were lower: one point six to two millimeters scored one, two to two point seven scored two, and more than two point seven scored three, with an extra point for inflamed mesenteric fat. A score of one or more had a sensitivity of eighty-two percent in the ileum and eighty-five percent in the colon, and a score of three or more had a specificity of eighty-eight and ninety-two percent. Agreement between observers was only moderate, with kappa values around zero point four five, and the score still needs external validation. In adults, the International Bowel Ultrasound Segmental Activity Score, called IBUS-SAS, combines wall thickness, inflammatory fat, Doppler signal and stratification, and is endorsed by the 2025 European guideline. The 2023 pediatric consensus of the International Bowel Ultrasound Group describes how to perform the examination in children, including Doppler settings sensitive enough for small vessels.

The most exciting use of ultrasound is early response. In the Super Sonic study, Dolinger and colleagues scanned forty-four children with ileal Crohn's disease starting biologic therapy. Sixty-six percent reached endoscopic remission. A fall of eighteen percent or more in terminal ileal wall thickness at week eight predicted remission with an area under the receiver operating characteristic curve of zero point nine nine, with a sensitivity of one hundred percent, a specificity of ninety-three percent and a negative predictive value of one hundred percent. By comparison, a fall in the Pediatric Crohn's Disease Activity Index had an area under the curve of only zero point six seven, and a fall in C-reactive protein only zero point four nine, no better than a coin toss. The study was small and limited to ileal disease, so one should confirm before changing therapy on ultrasound alone, but it shows that a bedside scan at week eight can tell the family something that blood tests cannot.

Part three. Magnetic resonance enterography.

Magnetic resonance enterography, often shortened to MR enterography, images the small bowel after the child drinks a large volume of oral contrast to distend it. In a 2013 systematic review by Giles and colleagues of eleven studies and 496 children, it detected active terminal ileal Crohn's disease with a pooled sensitivity of eighty-four percent and a specificity of ninety-seven percent. It is the test that maps the whole small bowel at diagnosis, finds skip lesions and strictures above the reach of the colonoscope, and, with pelvic sequences, shows perianal fistulas and abscesses.

The price is paid by the child. The protocol of the Society for Pediatric Radiology asks for a fast of four to six hours and an oral contrast volume scaled to body size, between 450 and 1,350 milliliters, taken over forty-five to sixty minutes, often with glucagon to stop peristalsis. Children younger than about seven usually need sedation or anesthesia. Intravenous gadolinium adds information about enhancement, but it also adds a needle and a contrast agent. In 2017 the United States Food and Drug Administration warned that all gadolinium agents leave traces in the brain and other tissues, more with the older linear agents than with macrocyclic agents, although no harmful effect has been shown, and advised that contrast be used only when necessary. Can it be left out? Lanier and colleagues studied fifty-two children in 2018 and found that adding contrast did not change diagnostic performance for luminal disease, but the scan without contrast missed three of five cases of perianal penetrating disease. So gadolinium can often be omitted for the bowel, but should be given when perianal or penetrating disease is suspected.

This is where the ImageKids study matters. Focht and colleagues followed 240 children prospectively, with a repeat MR enterography at eighteen months, to build a pediatric index. They started with thirty candidate features and, on 159 scans, reduced them to five: wall thickness, wall diffusion on diffusion weighted imaging, ulcerations, mesenteric edema, and

the comb sign. The resulting Pediatric Inflammatory Crohn's Magnetic Resonance Enterography Index, called PICMI, correlated with the global judgment of expert radiologists with a coefficient of zero point eight five, and with the endoscopic score with a coefficient of zero point six three. Agreement between radiologists was high, with an intraclass correlation of zero point eight four, and test retest reliability was zero point eight one. It responded well to change, with an area under the curve of zero point nine six across 116 repeat scans. A score of ten or less defines transmural healing, and a fall of more than twenty points defines response. Most important for children, PICMI needs no intravenous gadolinium and no rectal enema. Compare this with the adult index called MaRIA, developed by Rimola and colleagues in 2009, which detected active disease with a sensitivity of zero point eight one and a specificity of zero point eight nine but requires gadolinium, because it uses relative contrast enhancement. For a child, the practical message is to choose the index that avoids a needle: PICMI on magnetic resonance and wall thickness on ultrasound track the same inflammation that MaRIA measures with contrast.

Part four. Computed tomography and radiation.

Computed tomography gives superb pictures quickly, without sedation. Its cost is ionizing radiation, and the evidence that this matters in children is now strong. In the British cohort of Pearce and colleagues, about 178,000 young people who had CT scans were followed. Compared with a bone marrow dose under five milligray, a cumulative dose of thirty milligray or more carried a relative risk of leukemia of about three point two, and a brain dose of fifty to seventy-four milligray carried a relative risk of brain tumor of about two point eight. In absolute terms, the authors estimated about one extra leukemia and one extra brain tumor per ten thousand head scans in children under ten, over ten years. The Australian record linkage study of Mathews and colleagues covered 10.9 million people, of whom 680,211 had a CT scan in childhood or adolescence. Overall cancer incidence was twenty-four percent higher in the exposed group, with the risk rising with each additional scan, and the absolute excess was about one extra cancer for every 1,800 scans. The European EPI-CT consortium then studied 658,752 people from nine countries and found an excess relative risk of brain cancer of about one point three per hundred milligray, and in a second analysis of 948,174 people found an excess relative risk of blood cancers of about two per hundred milligray. With a typical modern scan delivering about eight milligray to the marrow, that translates to about one to two excess blood cancers per ten thousand children scanned, over about twelve years. These cohorts used lag periods to limit the problem of reverse causation, in which a scan is done because an undiagnosed cancer is already causing symptoms.

Why does this matter so much in inflammatory bowel disease? Because Crohn's disease is lifelong and is imaged again and again. A CT enterography in a child delivers between three

and ten millisieverts, while ultrasound and magnetic resonance deliver none. Each scan adds a small, cumulative, permanent risk. CT is therefore justified for abdominal trauma, for an unstable child with suspected perforation, and to guide percutaneous drainage of an abscess. It is debatable for an acute flare when magnetic resonance is unavailable, or for an out of hours abscess question when ultrasound is not diagnostic. It should be avoided for the routine diagnosis and monitoring of pediatric inflammatory bowel disease, and for suspected appendicitis before ultrasound has been tried.

Part five. Malrotation and the infant with bilious vomiting.

Malrotation means that the midgut did not complete its normal rotation and fixation in fetal life. The duodenojejunal flexure, the point where the duodenum turns into the jejunum and is held by the ligament of Treitz, ends up in the wrong place, and the mesentery is attached along a narrow stalk around the superior mesenteric artery. That narrow stalk can twist, producing midgut volvulus, which strangles the entire small bowel within hours. The classic sign is bilious, meaning green, vomiting in a baby. Not every such baby has a surgical problem: in one British series of term neonates with bilious vomiting, fourteen point six percent had a surgical cause and three point three percent had malrotation, while earlier tertiary series reported surgical causes in twenty to fifty-one percent. But because volvulus can kill or destroy the gut so quickly, every green vomit is treated as a surgical clock.

The upper gastrointestinal contrast study has long been the gold standard. It shows the position of the duodenojejunal flexure. Normally the flexure lies to the left of the left vertebral pedicle, at the level of the duodenal bulb, and posterior on the lateral view, because the duodenum is fixed behind the peritoneum. In the series by Sizemore and colleagues of 166 children, 163 of whom had surgically verified malrotation, the contrast study had a sensitivity of ninety-six percent, with two false positives and seven false negatives, and it showed thirty of thirty-eight cases of volvulus. False positives arise in neonates with a mobile duodenum, when bowel or a mass displaces the duodenum, or when the stomach is distended by air or a nasogastric tube. False negatives arise when a slightly low flexure is called normal, when the position of the jejunum is over-trusted, or when the study is stopped too early. Volvulus itself appears as a corkscrew duodenum or an abrupt beak-like obstruction.

Ultrasound has caught up. Older studies relied on the relationship of the superior mesenteric artery and vein, and found sensitivities of only seventy and eighty-seven percent, with failed examinations in up to one in six. The modern technique adds the whirlpool sign, a clockwise twist of the vein and mesentery around the artery, and follows the third part of the duodenum to confirm that it crosses behind the artery, often after giving fluid to fill it. In the 2021 meta-analysis by Nguyen and colleagues, which pooled seventeen studies of 2,257 children, ultrasound had a sensitivity of ninety-four percent and a specificity of one hundred percent,

with better performance in neonates, when fluid was given first, and in malrotation without volvulus. Several units, including Great Ormond Street in London, now use an ultrasound first pathway. No society guideline yet settles the question, so the safe rule is this: a whirlpool sign means theatre, but a normal vessel relationship does not exclude malrotation, and an equivocal ultrasound in a baby with bilious vomiting goes straight to an urgent contrast study or to the surgeon.

The pathway is simple. Recognize that the vomit is green rather than yellow and assume malrotation with volvulus. Stop feeds, decompress the stomach with a nasogastric tube, start fluids and call the surgeon now. Image with ultrasound if an experienced operator is immediately available, otherwise with an urgent contrast study. Then decide: a whirlpool sign or an abnormal flexure means theatre. One North American pathway sets a target of under one hour from imaging result to theatre.

Part six. Intussusception, appendicitis and Hirschsprung disease.

For intussusception, ultrasound is the test. In the series of 814 examinations by Hryhorczuk and Strouse, radiology performed ultrasound had a sensitivity of ninety-seven point nine percent, a specificity of ninety-seven point eight percent and a negative predictive value of ninety-nine point seven percent. Point of care ultrasound by emergency physicians, in a 2020 meta-analysis by Lin-Martore and colleagues of 1,303 children, had a sensitivity of about ninety-five percent and a specificity of about ninety-nine percent. For appendicitis, the 2006 meta-analysis by Doria and colleagues found that ultrasound in children had a sensitivity of eighty-eight percent and a specificity of ninety-four percent, against ninety-four and ninety-five percent for CT. CT gains a few percentage points at the cost of dose, which is why ultrasound comes first.

Hirschsprung disease is a different story. In the systematic review by de Lorijn and colleagues, the contrast enema had a sensitivity of only seventy percent and a specificity of eighty-three percent, compared with ninety-one and ninety-four percent for anorectal manometry and ninety-three and ninety-eight percent for rectal suction biopsy. And the radiological transition zone, the point where narrow bowel widens on the enema, matched the true level of aganglionosis at surgery in only half of the children in one series. The enema is a road map, not a diagnosis, and the extent of disease is a biopsy question.

Part seven. What the guidelines say.

On small bowel imaging at diagnosis, North America and Europe are aligned. The revised Porto criteria of ESPGHAN recommend small bowel imaging, by magnetic resonance enterography or capsule endoscopy, in all children with suspected inflammatory bowel disease except those with typical ulcerative colitis, and they call magnetic resonance the imaging

modality of choice. The 2025 American College of Gastroenterology guideline also asks for small bowel imaging at diagnosis. On the preferred cross-sectional test, they are again aligned: the American guideline calls CT enterography comparable to magnetic resonance but prefers magnetic resonance in patients under thirty-five, and European pediatric guidance names magnetic resonance as the modality of choice in children.

The differences are about ultrasound and CT. The 2025 joint European guideline of ECCO with the radiology, pathology and bowel ultrasound societies promotes intestinal ultrasound to a first line tool for both diagnosis and monitoring. In North America, the American College of Radiology rates ultrasound only as may be appropriate for the initial workup of pediatric Crohn's disease, and the American College of Gastroenterology calls it an adjunct. The material difference is radiation: the 2022 American College of Radiology criteria still rate CT enterography as usually appropriate for the pediatric Crohn's workup, alongside magnetic resonance, while European pediatric practice reserves CT for emergencies. Two gaps remain on both sides. No society has yet endorsed a pediatric activity index such as PICMI in its guideline text, and no society guideline settles whether ultrasound may replace the contrast study for malrotation.

Part eight. Traps and pearls.

The first trap is using adult numbers: a three millimeter cut-off in a four-year-old will call an inflamed ileum normal; in children, think of about two point two to two point seven millimeters. The second is reading diffusion restriction alone, which overcalls activity because normal collapsed jejunum restricts. The third is the reassuring normal vessel relationship on ultrasound, which does not exclude malrotation, leading to an equivocal scan that is not escalated and a delay to theatre. The fourth is using the enema to plan the extent of a Hirschsprung resection, when only the biopsy can show where ganglion cells begin. Above all of these sits one rule: a negative scan never overrides a bilious vomit, a pale stool or a rising fecal calprotectin. Image again, scope, or call the surgeon.

The pearls are worth repeating. The normal pediatric bowel wall is no thicker than one point nine millimeters, and the adult three millimeter cut-off missed sixty percent of inflamed ilea in very early onset disease. Ultrasound rules disease in well, with a specificity of ninety to one hundred percent, but rules it out less well. A fall of eighteen percent or more in ileal wall thickness at week eight predicted remission almost perfectly, where C-reactive protein did not. PICMI has five items, needs no gadolinium and no enema, and defines healing as ten or less and response as a fall of more than twenty. Magnetic resonance detects active ileal Crohn's disease with a sensitivity of eighty-four percent and a specificity of ninety-seven percent, and gadolinium should be added for perianal disease. CT carries a small cumulative cancer risk of about one to two extra blood cancers per ten thousand children scanned. Ultrasound for

malrotation is ninety-four percent sensitive and one hundred percent specific in expert hands, but an equivocal scan still needs a contrast study. And ultrasound for intussusception is about ninety-eight percent sensitive, with a negative predictive value near one hundred percent.

Part nine. What to do differently in clinic.

Return to the four children. The fourteen-year-old starting an anti tumor necrosis factor drug with a baseline ileal wall of five point five millimeters should have a repeat ultrasound at week eight. If the wall has fallen to four millimeters, a fall of twenty-seven percent, he is very likely heading for endoscopic remission, even if his C-reactive protein has not moved, and a magnetic resonance enterography scored with PICMI at about twelve months can confirm transmural healing. The twelve-year-old with suspected Crohn's disease needs a small bowel map, as the Porto criteria require: ileocolonoscopy plus magnetic resonance enterography, a baseline ultrasound for later comparison, and no CT. The four-year-old with bloody diarrhea and a colon wall of two point three millimeters does not have a normal scan; for her age, that wall is abnormal. She needs endoscopy, ultrasound for follow-up, and magnetic resonance only if it is truly needed, because it would require anesthesia. And the ten-day-old with bilious vomiting and a whirlpool sign on ultrasound goes to theatre now, with no further imaging. If the ultrasound had been equivocal, the next step would be an urgent contrast study, not reassurance. Finally, a teenager with Crohn's disease who arrives at night with fever, a tender mass and unstable vital signs should have ultrasound first, and CT only if he is unstable or needs drainage.

The whole chapter can be held in two sentences. Measure a child's bowel with a child's ruler, use ultrasound as the first and repeated look, and use magnetic resonance with PICMI as the map and the healing endpoint, keeping CT for emergencies. For green vomit, the clock is running: ultrasound may show the twist, but if it is not certain, the contrast study or the surgeon decides.

Questions to test yourself.

Question one. What is the upper limit of normal bowel wall thickness in healthy children, and why does the adult cut-off fail? Answer: about one point nine millimeters for both small bowel and colon; the adult cut-off of three millimeters sits well above the pediatric range and missed sixty percent of inflamed ilea in very early onset disease.

Question two. What ultrasound change at week eight predicted endoscopic remission in the Super Sonic study? Answer: a fall of eighteen percent or more in terminal ileal wall thickness, with an area under the curve of zero point nine nine, compared with zero point four nine for the fall in C-reactive protein.

Question three. What are the five items of PICMI, and what are its thresholds? Answer: wall thickness, wall diffusion, ulcerations, mesenteric edema and the comb sign; a score of ten or less defines transmural healing and a fall of more than twenty points defines response, and no gadolinium or enema is required.

Question four. When should gadolinium be given for magnetic resonance enterography in a child? Answer: when perianal or penetrating disease is suspected, because a non contrast study missed three of five such cases.

Question five. What is the approximate excess risk of blood cancer after CT in childhood? Answer: about one to two extra cases per ten thousand children scanned, at a typical modern dose of about eight milligray, over about twelve years.

Question six. How accurate are the contrast study and ultrasound for malrotation? Answer: the contrast study had a sensitivity of ninety-six percent in the Sizemore series, and ultrasound had a pooled sensitivity of ninety-four percent and a specificity of one hundred percent in the 2021 meta-analysis; an equivocal ultrasound still needs a contrast study.

Question seven. Why should the contrast enema not be used to define the extent of Hirschsprung disease? Answer: because its sensitivity is only about seventy percent and the radiological transition zone matched the true level of aganglionosis in only half of children; rectal suction biopsy, with a sensitivity of ninety-three and specificity of ninety-eight percent, is the diagnostic test.

End of chapter.